328613(28)

BE (6th Semester)

Examination, April-May, 2016

(Old Scheme)

Advanced Microprocessor and Interfacing

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Time Allowed : 3 hours *Maximum Marks* : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory. Attempt any two parts from (*b*), (*c*) and (*d*) of each question.

(*ii*) Draw diagram and symbols wherever required.

(*iii*) The figures in the right-hand margin indicate marks.

1. (*a*) The CPU of the 8086 & 8088 microprocessor is divided into an _____ unit and a _____ unit. 2

(*b*) What do you mean by pipelined architecture? Draw internal block diagram of 8086 and discuss only general purpose registers available for 8086. 7

(*c*) What do you mean by addressing modes? Explain the physical address formation in different addressing modes with suitable examples. 7

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(Turn Over)

(*d*) Write a program to find out the largest no. from a given unordered array of 8 bits numbers stored in 100 locations. Store the largest number in a location labeled as largest number. 7

2. (*a*) Write the pin with pin no. of 8086 which is responsible for functioning the machine in minimum mode or maximum mode. 2

(*b*) From which address 8086 starts execution after reset? Draw and discuss a typical maximum mode 8086 System. 7

(*c*) Interface 16K bytes of EPROM memory to 8086 with starting address 10000H. 7

(*d*) What is the difference between hardware and software interrupt? Explain the interrupt response sequence of 8086? 7

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3. (*a*) What are the various operating modes of 8253? 2

(*b*) Draw and discuss the internal Block diagram of 8259 PIC. 7

(*c*) Explain different commands of 8279 in brief. 7

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(*d*) Design a programmable timer, using 8253 and 8086. Interface 8253 at an addrress 004017 for counter 0 and write the following ALPS. The 8086 and 8253 run at 6MHz and 1.5MHz respectively :

(*i*) To generate a square wave of period 1ms.

(*ii*) To interrupt the processor after 10ms. 7

4. (*a*) The extended BX register is addressed as _____. 2

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(*b*) Draw and discuss the register set of 80386. 7

(*c*) What do you mean by paging? What are its advantages and disadvantages? 7

(*d*) Explain the physical address formation in PVAM of 80386. 7

5. (*a*) What is the frequency range of 8087 NDP? 2

(*b*) Draw and discuss the architecture of 8089 I/O processor. 7

(*c*) How does the CPU differentiate the 8087 instruction from its own instruction? Draw and discuss the interface between 8086 and 8087. 7

(*d*) Write any 8086/8087 procedure to calculate the area of a circle. 7

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